

Consumer and Producer Surplus

**ECON 101: Introduction to
Microeconomics**

Prof. Alejandro Melo Ponce

The Demand and Supply Schedules as the marginal benefit and cost

- **Demand Schedule** The inverse demand represents the consumers' *willingness to pay* for a certain quantity of the good, that is, the maximum price that the consumers are willing to pay to buy a particular quantity of the good. This means that the demand curve also represents the **marginal benefit of consumption**.
- **Supply Schedule** The inverse supply represents the producers' *willingness to sell* a certain quantity of the good, that is, the minimum price that producers are willing to receive when selling a particular quantity of the good. This means that the supply curve also represents the **marginal cost of production**.
- Both curves show the **reservation value** of each participant in the market.



Example with discrete quantities: the used iPhone Market

Reservation Values of Buyers and Sellers in the iPhone Market

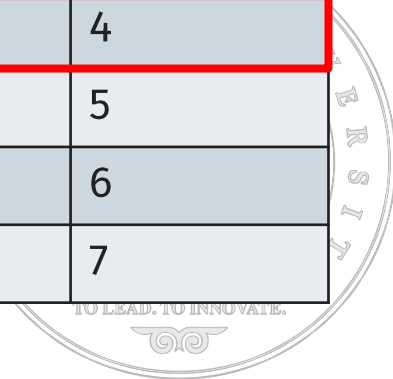
Buyers	Res. Value (\$)	Cum. Q	Sellers	Res. Value (\$)	Cum. Q
Madeline	\$70	1	Tom	\$10	1
Katie	\$60	2	Mary	\$20	2
Sean	\$50	3	Jeff	\$30	3
Dave	\$40	4	Phil	\$40	4
Ian	\$30	5	Adam	\$50	5
Kim	\$20	6	Matt	\$60	6
Ty	\$10	7	Fiona	\$70	7

Example with discrete quantities: the used iPhone Market

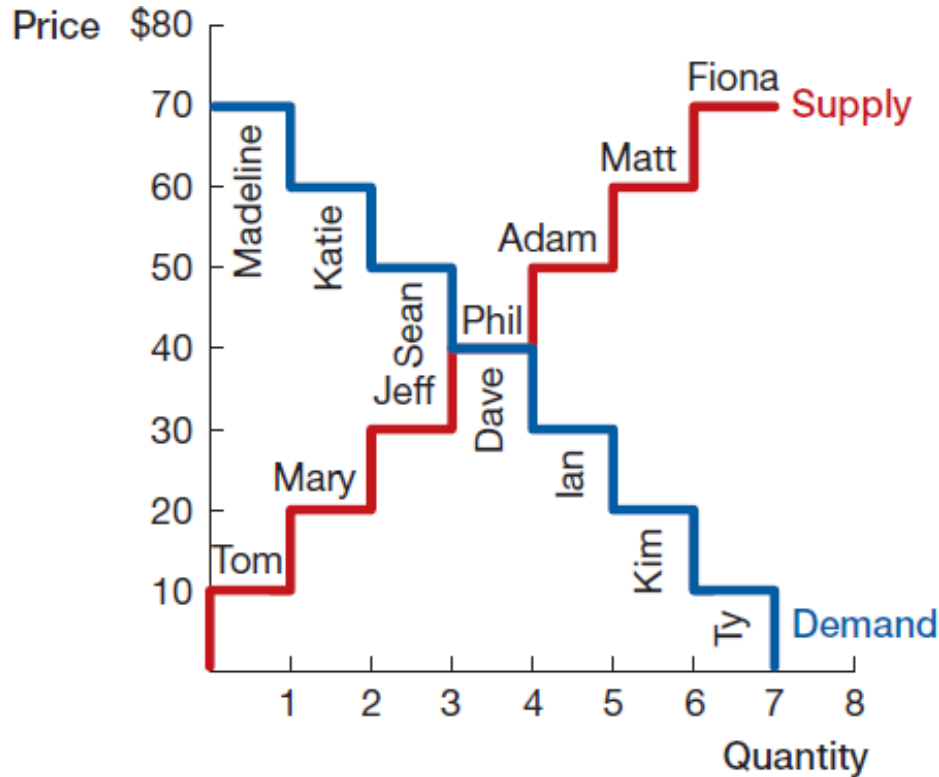
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At \$40 Market is in Equilibrium



Example with discrete quantities: the used iPhone Market



Example with discrete quantities: the used Iphone Market

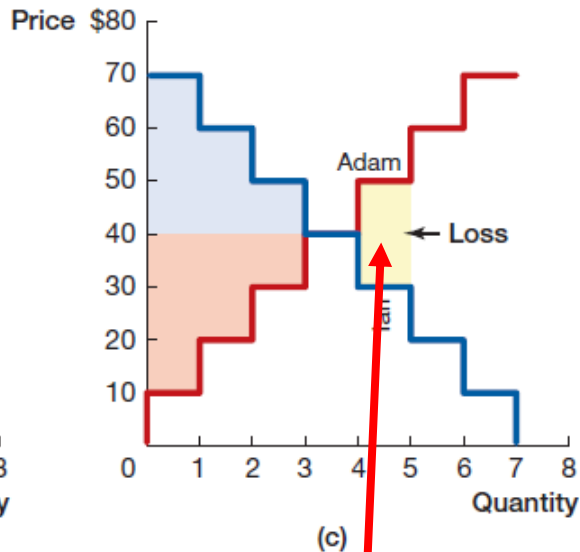
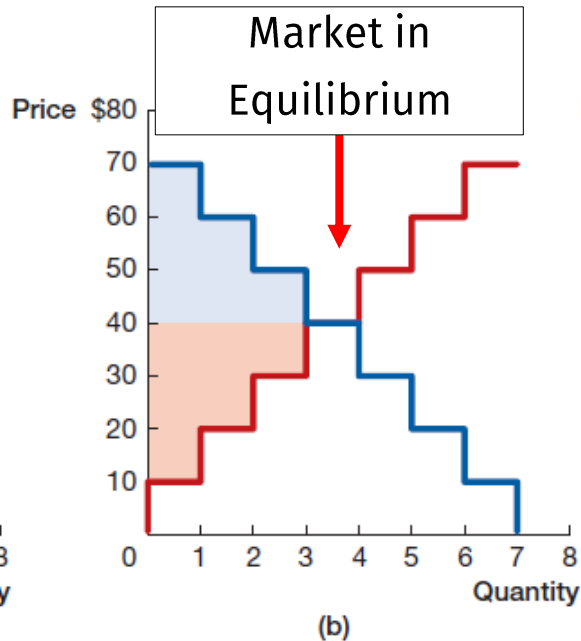
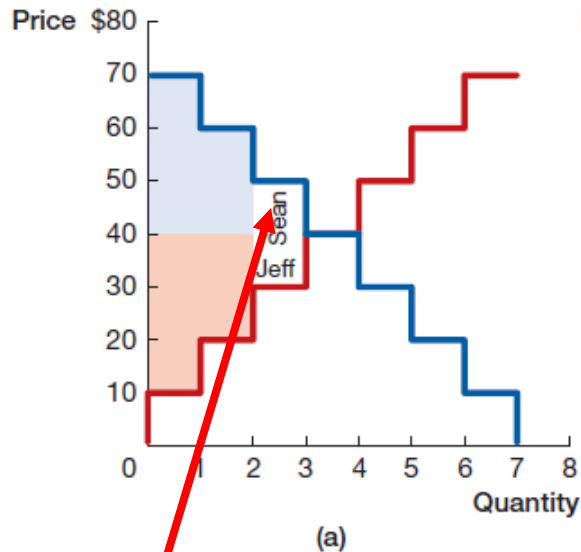
Reservation Values vs Price Paid

Buyers	Res. Value (\$)	Price Paid (\$)	Consumer Surplus (\$)	Sellers	Price Received (\$)	Res. Value (\$)	Producer Surplus	Cumulative Quantity
Madeline	\$70	\$40	\$30	Tom	\$40	\$10	\$30	1
Katie	\$60	\$40	\$20	Mary	\$40	\$20	\$20	2
Sean	\$50	\$40	\$10	Jeff	\$40	\$30	\$10	3
Dave	\$40	\$40	\$0	Phil	\$40	\$40	\$0	4
Ian	\$30			Adam		\$50		5
Kim	\$20			Matt		\$60		6
Ty	\$10			Fiona		\$70		7
Total	\$220	\$160	\$60	Total	\$160	\$100	\$60	Blank

Total surplus (Social Surplus)=Consumer Surplus + Producer Surplus



Efficiency and Social Surplus



Restrict
Quantity
Lose this area

We cannot improve the outcome
by forcing the price or the
quantity higher or lower.

Force Adam to sell at
price \$30 (his WTS \$50)
Lose this area

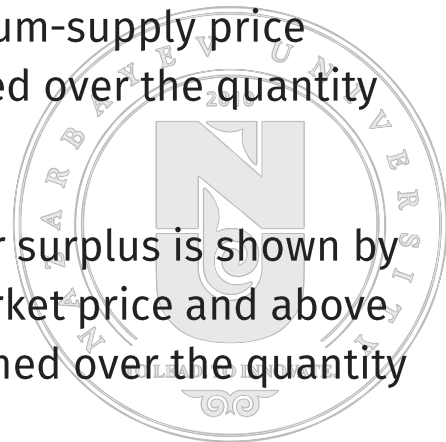
Consumer and Producer Surplus

Consumer Surplus

- **Consumer surplus** is the excess of the benefit received from a good over the amount paid for it.
- We can calculate consumer surplus as the marginal benefit (or value) of a good minus its price, summed over the quantity bought.
- It is measured by the area under the demand curve and above the price paid, up to the quantity bought.

Producer Surplus

- **Producer surplus** is the excess of the amount received from the sale of a good over the cost of producing it.
- We calculate it as the price received for a good minus the minimum-supply price (marginal cost), summed over the quantity sold.
- On a graph, producer surplus is shown by the area below the market price and above the supply curve, summed over the quantity sold.



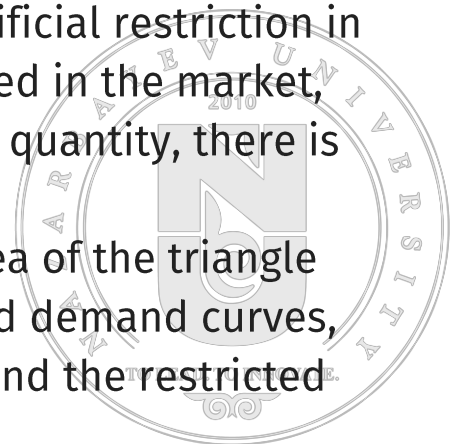
Social Surplus and Deadweight Loss

Social Surplus

- Social Surplus is the sum of consumer surplus and producer surplus
- It represents the total gains from participating in a competitive market.
- We calculate it as the area of triangle formed by the supply and demand curves and their intersection at the market equilibrium.

Deadweight Loss

- The Deadweight loss is the cost to society created by market inefficiency.
- It is created when the socially optimal quantity of a good is not traded in the market.
- Whenever there is an artificial restriction in the quantity that is traded in the market, i.e. an out of equilibrium quantity, there is a deadweight loss.
- We calculate it as the area of the triangle formed by the supply and demand curves, the market equilibrium and the restricted quantity.



① $Q_D = Q_D(P, Det^D)$
 ② $Q_S = Q_S(P, Det^S)$

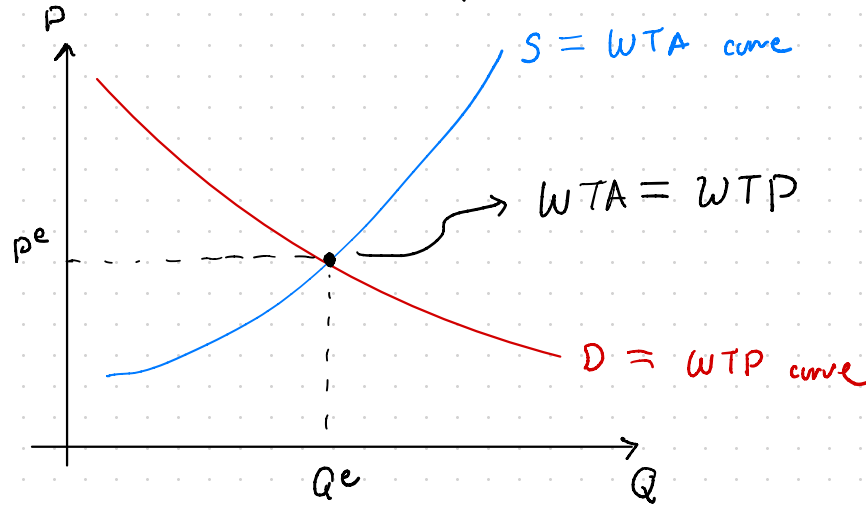
Given a price P and respectively Det^D, Det^S ,
 How each participant (consumers/producers) are
 willing to buy or sell in the market

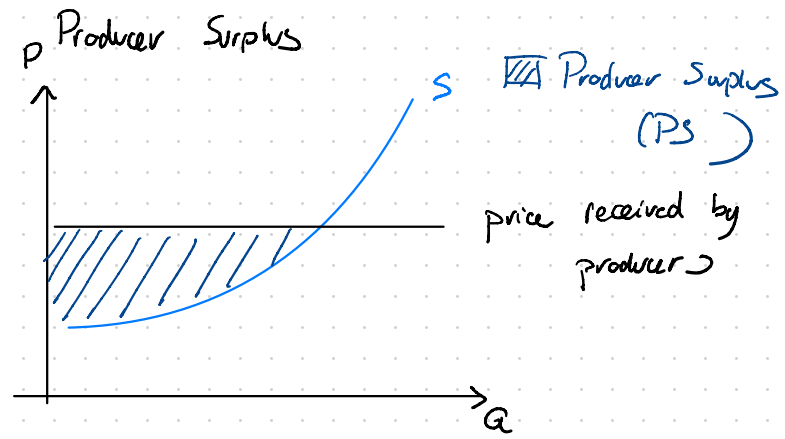
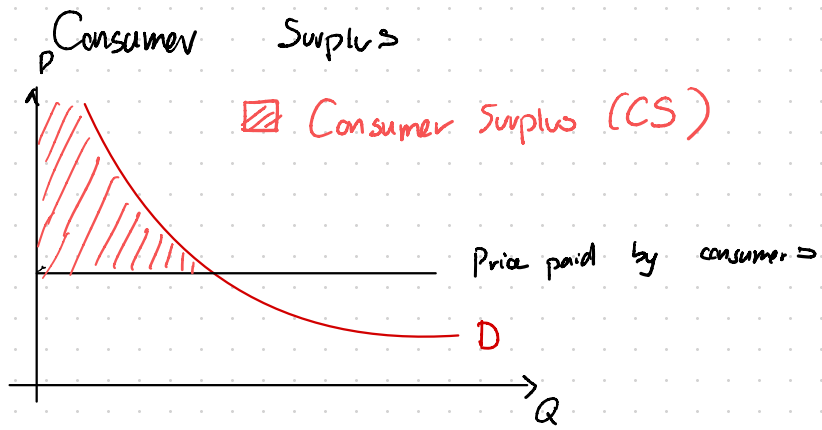
↓

Inverse schedules ; for fixed levels of Det^D, Det^S

① $P_D = P_D(Q_D, Det^D)$ How much the consumer is willing to pay. WTP

② $P_S = P_S(Q_S, Det^S)$ The minimum price the seller is willing to accept WTA





$$\text{Social surplus (TS)} = \text{CS} + \text{PS}$$

Linear demand & supply schedules

$$Q_D = a - bP$$

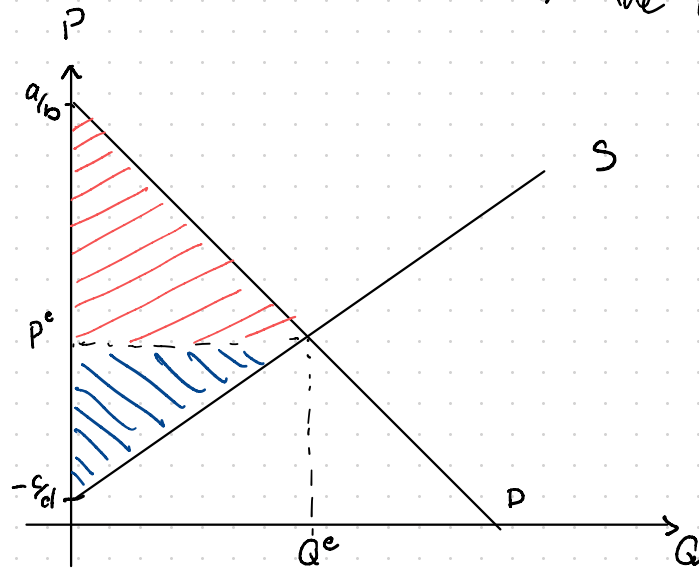
$$Q_S = c + dP$$

$$a, b, d > 0, \quad c$$

a, d, c are placeholders
for the determinants

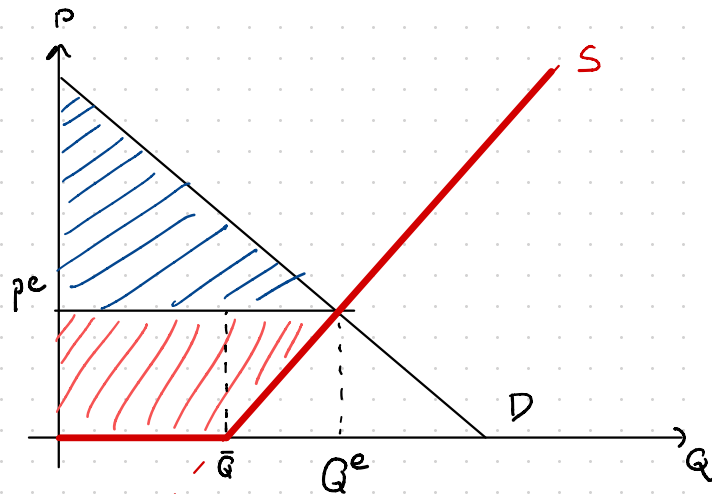
$$p^e = \frac{a-c}{b+d}$$

$$Q^e = \frac{ad-bc}{b+d}$$



$$CS = \frac{Q_e \left(\frac{a}{b} - p^e \right)}{2}$$

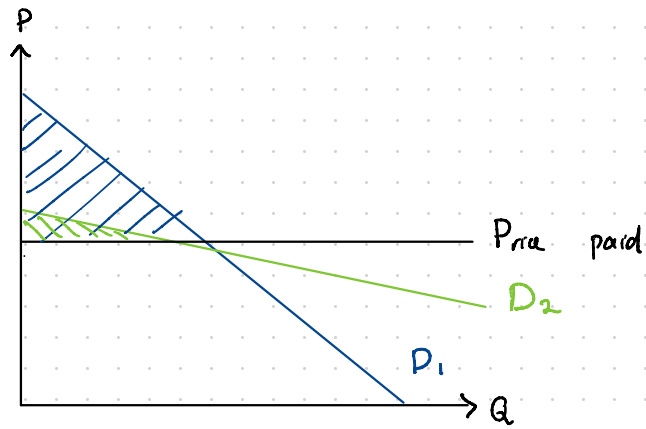
$$PS = \frac{Q_e \left(p^e - \left(-\frac{c}{d} \right) \right)}{2} = \frac{Q_e \left(p^e + \frac{c}{d} \right)}{2}$$



$$CS = \text{blue shaded area}$$

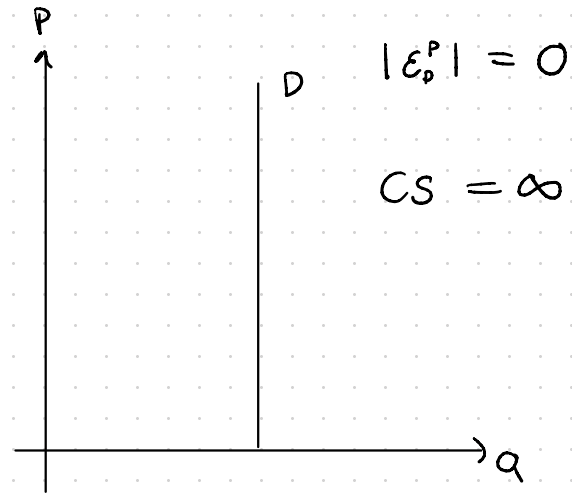
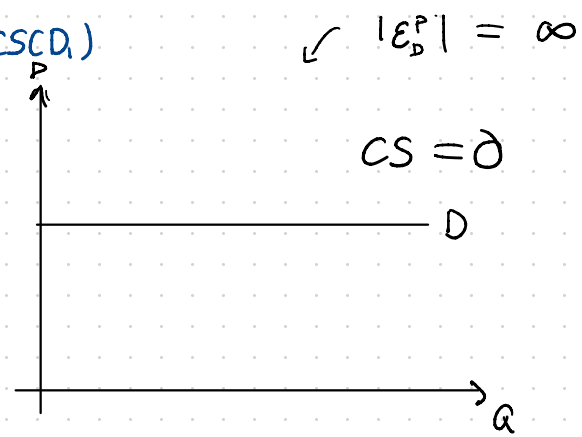
$$PS = \text{red shaded area} = \bar{Q} \times p^e + \frac{(Q^e - \bar{Q}) \times p^e}{2}$$

Those sellers are willing to offer $\bar{Q}$ at price $p=0$

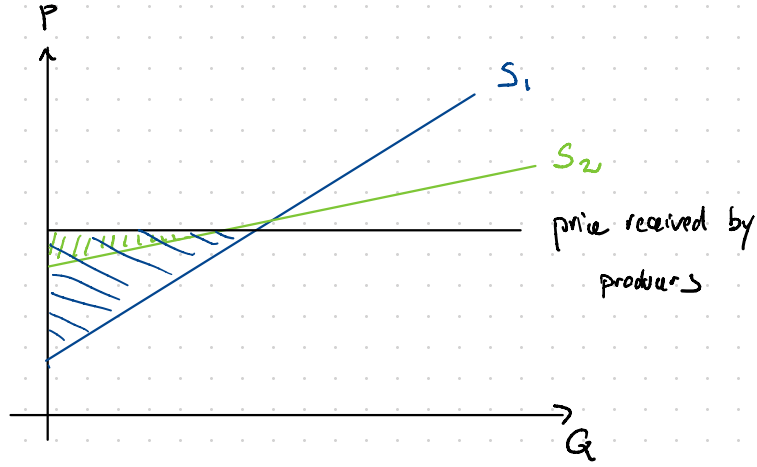


Consumer Surplus & Elasticity (Price)
of demand

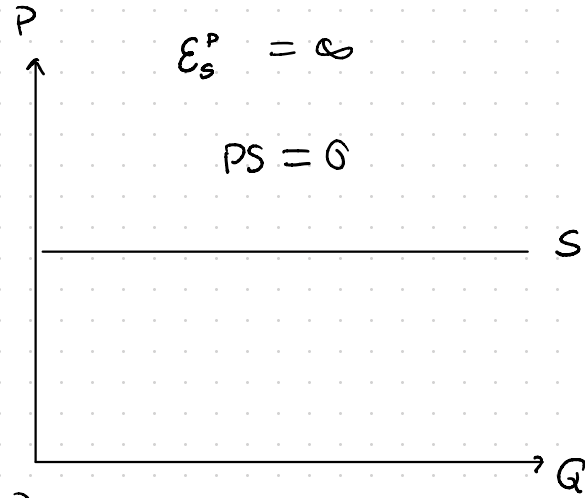
$\Rightarrow$



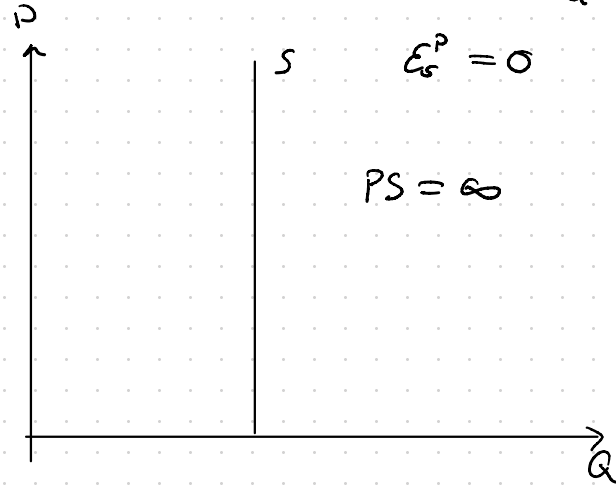
Elasticity of Supply (price) & Producer Surplus



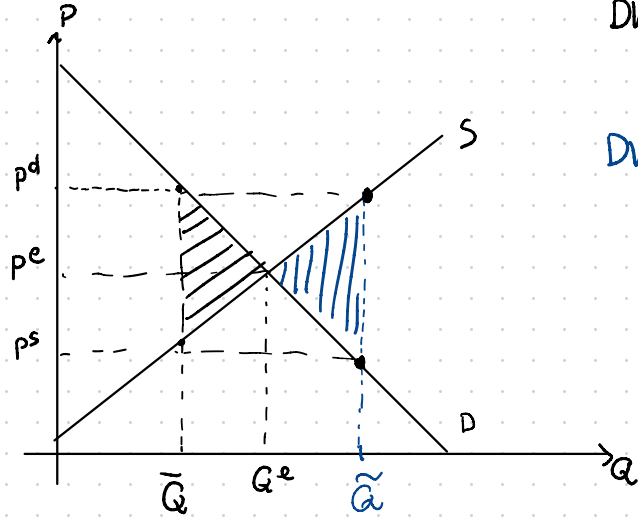
$\Rightarrow$



$\Rightarrow$



Deadweight Loss



$$\bar{Q} < Q^e$$

$$DWL = \text{shaded area} = \frac{(Q^e - \bar{Q})(P^d - P^s)}{2}$$

$$DWL = \text{shaded area} = \frac{(\tilde{Q} - Q^e)(P^d - P^s)}{2}$$



End of this lecture.

Chapter 5 of Parkin contains some of the material in this lecture.